Deeksha Dangwal

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Research Focus

I am a systems researcher working on AI safety. My background is in computer architecture and hardware security, and I study where safety mechanisms should be enforced in the training stack. My current work is on compute verification and on training-time safety monitoring. My work in computer architecture, privacy, and secure computation has appeared in ISCA, ASPLOS, PLDI, FPL, BMVC, and TMLR. My SoK on hardware-software-security codesign was cited in the *US National Strategy on Microelectronics Research*, I am a Rising Stars in EECS alumna, and trace wringing was an IEEE Micro Top Pick.

Education

2022 **Ph.D. in Computer Science**, University of California, Santa Barbara
Thesis: *A System-level Framework for Privacy*. Advisor: Timothy Sherwood

2016 **M.S. in Electrical and Computer Engineering**,
University of California, Santa Barbara

2014 **B.E. in Instrumentation and Electronics**, Ramaiah Institute of Technology, Bangalore

Appointments

2026– **Fellow**, MATS Research, Berkeley, CA. Mentor: James Petrie (Future of Life Institute)

- *Compute verification*: designing and empirically evaluating software-based compute verification protocols (memory-state challenges, subsampled zero-knowledge proofs) that bound undeclared GPU workloads without trusted hardware

2026– **Research Fellow**, ERA:AI Fellowship in Technical AI Governance, Cambridge, UK. Mentor: Alan Cooney (Google DeepMind)

- *Training-time safety monitoring*: asks whether a fine-tuning job that removes a model's safety training can be detected while it is still running. The monitor runs below the privilege level of the party running the job, outside that party's control

2025 **Consultant**, Office of the CTO, The Allen Institute, Seattle, WA

- *Neural tracing*: co-developed neural tracing for mechanistic interpretability of LLMs and bio-foundation models. Patent application filed [P2]

2022–25 **Research Scientist**, Reality Labs Research, Meta

- *Distributed intelligence and on-device AI*: prototyped gesture interaction for smart glasses (data collection, model training, compression, and demo) on an SoC with custom and off-the-shelf accelerators. Through HW/SW codesign and energy-aware scheduling, the pipeline improved latency and power by 93% and 74%

- *Performance and power modeling*: built a Python-based system-level power and performance modeling tool for Project Aria and other wearable devices using discrete-event simulation, calibrated against measurements on prototype devices running machine perception pipelines (eye tracking, hand and gesture tracking, localization and mapping)
- *Hardware-software-security codesign in extended reality*: developed a security and privacy research program for mixed and augmented reality, establishing threat models for codec avatars, privately crowdsourced data collection for world modeling, and other XR applications [W3, C5, C4, J4, A1]

2015–21 **Graduate Student Researcher**, ArchLab, University of California, Santa Barbara

- *Trace wringing*: developed trace wringing for safer program behavior sharing, achieving privacy through lossy compression and information flow tracking with verifiable leakage bounds [C3, C6]; recognized as an IEEE Micro Top Pick [J2]
- *Agile hardware development with PyRTL*: co-developed PyRTL, a Pythonic hardware development toolkit for rapid prototyping and agile hardware design [C1, J3]
- *Architecture modeling with Charm*: worked on Charm, a domain-specific language for high-level architecture modeling [C2, J1]
- *OpenTPU*: led development of an open-source tensor processing unit in PyRTL

Research Internships

2020–21 **Research Scientist Intern**, Reality Labs Research, Meta

- Implemented a novel reverse engineering attack on local feature descriptors, surpassing state-of-the-art reconstruction accuracy for user image recovery; established the first privacy threat model for feature descriptor sharing in AR systems; developed privacy-preserving mitigations and studied their effect on downstream vision performance [C4]

2018 **Research Intern**, Microsoft Research

- Implemented parameterizable architecture-aware machine learning graph primitives for custom hardware instructions on the Brainwave Neural Processing Unit; built tools for automatic lifting of hardware instructions to high-level graph primitives; designed a computational pipeline for NPU model decompilation with accuracy verification [P1]

2016 **Research Assistant**, Oracle Labs

- Established a testing environment for measuring throughput of the RAPID Data Processing Unit network, a bandwidth-optimized big data computation architecture; implemented network congestion tests using hardware RPC acceleration

Awards and Honors

2024 Paper cited in the *United States National Strategy on Microelectronics Research* [W3]
2021 Runner-up, UCSB Grad Slam Competition
2020 Rising Stars in EECS, UC Berkeley
2020 IEEE Micro Top Pick [J2]
2020 Fiona and Michael Goodchild Graduate Mentoring Award, UCSB Graduate Division
2020 Outstanding Graduate Student Award, Department of Computer Science, UCSB

Publications

See my Google Scholar profile for citation records. In computer architecture and systems, top-tier conferences are the primary archival venues and are comparable to journals in other disciplines in selectivity and impact. [C] conference, [J] journal or magazine, [W] workshop, [A] preprints and reports, [P] patents.

Conference Publications

[C6] **D. Dangwal**, Z. Zhang, J. Crandall, T. Sherwood. "Context-Aware Privacy-Optimizing Address Tracing". *IEEE International Symposium on Secure and Private Execution Environment Design (SEED)*, 2021. [link]

[C5] M. Cowan, **D. Dangwal**, A. Alaghi, C. Trippel, V. T. Lee, B. Reagen. "Porcupine: A Synthesizing Compiler for Vectorized Homomorphic Encryption". *ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, 2021. [link]

[C4] **D. Dangwal**, V. T. Lee, H. J. Kim, T. Shen, M. Cowan, R. Shah, C. Trippel, B. Reagen, T. Sherwood, V. Balntas, A. Alaghi, E. Ilg. "Mitigating Reverse Engineering Attacks on Local Feature Descriptors". *British Machine Vision Conference (BMVC)*, 2021. [link]

[C3] **D. Dangwal**, W. Cui, J. McMahan, T. Sherwood. "Safer Program Behavior Sharing through Trace Wringing". *International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2019. [link]
Selected as an IEEE Micro Top Pick [J2].

[C2] W. Cui, Y. Ding, **D. Dangwal**, A. Holmes, J. McMahan, A. JavadiAbhari, G. Tzimpragos, F. Chong, T. Sherwood. "Charm: A Language for Closed-form High-level Architecture Modeling". *International Symposium on Computer Architecture (ISCA)*, 2018. [link]

[C1] J. Clow, G. Tzimpragos, **D. Dangwal**, S. Guo, J. McMahan, T. Sherwood. "A Pythonic Approach for Rapid Hardware Prototyping and Instrumentation". *International Conference on Field-Programmable Logic and Applications (FPL)*, 2017. [link]

Journal and Magazine Publications

[J4] S. Q. Zhang, Z. Li, C. Guo, S. Mahloujifar, **D. Dangwal**, E. Suh, B. D. Salvo, C. Liu. "Unlocking Visual Secrets: Inverting Features with Diffusion Priors for Image Reconstruction". *Transactions on Machine Learning Research (TMLR)*, 2025. [link]

[J3] **D. Dangwal**, G. Tzimpragos, T. Sherwood. "Agile Hardware Development and Instrumentation with PyRTL". *IEEE Micro, Special Issue on Agile and Open Source Hardware*, 2020. [link]

[J2] **D. Dangwal**, W. Cui, J. McMahan, T. Sherwood. "Trace Wringing for Program Trace Privacy". *IEEE Micro, Top Picks from the Computer Architecture Conferences*, 2020. [link]
IEEE Micro Top Pick.

[J1] W. Cui, G. Tzimpragos, Y. Tao, J. McMahan, **D. Dangwal**, N. Tsiskaridze, G. Michelogiannakis, D. Vasudevan, T. Sherwood. "Language Support for Navigating Architecture Design in Closed Form". *ACM Journal on Emerging Technologies in Computing Systems (JETC)*, 2019. [link]

Workshop Publications

[W4] **D. Dangwal**, A. Rajagopal. "The Need for Computational Pluralism". *Workshop on Ethical Systems and Architecture Design (HotEthics)*, 2026.

[W3] **D. Dangwal**, M. Cowan, A. Alaghi, V. Lee, B. Reagen, C. Trippel. "SoK: Opportunities for Software-Hardware-Security Codesign for Next Generation Secure Computing". *Hardware and Architectural Support for Security and Privacy (HASP)*, 2020. [link]
Cited in the United States National Strategy on Microelectronics Research, 2024.

[W2] D. Aboye, D. Kupsh, M. Lim, J. Mai, **D. Dangwal**, D. Mirza, T. Sherwood. "PyRTLMatrix: An Object-Oriented Hardware Design Pattern for Prototyping ML Accelerators". *Workshop on Energy Efficient Machine Learning and Cognitive Computing for Embedded Applications (EMC2)*, 2019. [link]

[W1] D. Mirza, **D. Dangwal**, T. Sherwood. "PyRTL in Early Undergraduate Research". *Workshop on Computer Architecture Education (WCAE)*, 2019. [link]

Preprints and Reports

[A2] "Mechanism Design for Improving Hardware Security". *Computing Community Consortium (CCC) Workshop Report*, August 2022. [link]

[A1] **D. Dangwal**, V. T. Lee, H. J. Kim, T. Shen, M. Cowan, R. Shah, C. Trippel, B. Reagen, T. Sherwood, V. Balntas, A. Alaghi, E. Ilg. "Analysis and Mitigations of Reverse Engineering Attacks on Local Feature Descriptors". arXiv preprint, May 2021. [link]

Patents

[P2] A. Rajagopal, G. H. Huynh, **D. Dangwal**. "Model Optimization and Data Analysis Using Neural Traces". Provisional patent filed September 2025.

[P1] J. Fowers, D. Lo, **D. Dangwal**. "Deriving a Concordant Software Neural Network Layer from a Quantized Firmware Neural Network Layer". US Patent 11,556,764, Microsoft Technology Licensing LLC, 2023. [link]

Invited Talks and Seminars

2025 *Performance, Power, and Privacy: Codesign Strategies for Always-On Wearables*. Washington State University, Pullman

2025 *Privacy and Security in the Age of Metaverse: A Case for Data Minimization and On-device AI*. Washington State University, Everett

2021 *A System-Level Framework for Privacy: Applications of Wringing in AR/VR*.
Reality Labs Research, Meta

2021 *A System-Level Framework for Privacy*. Arizona State University

2021 *A System-Level Framework for Privacy*. Pennsylvania State University

2021 *Privacy through Wringing*. UCSB Grad Slam Final

2019 *Agile Hardware Development and Instrumentation with PyRTL*. Agile-RTL Workshop, UC Santa Barbara

2019 *OpenTPU: Prototyping ML Accelerators with PyRTL*.
IEEE Space Computing Conference, Caltech

Teaching and Mentorship

2026 **Mentor**, SPAR (Supervised Program for Alignment Research), Fall 2026
Project: *Constraint Drift Through Delegation Hierarchies*, co-mentored with Abhejit Rajagopal (Allen Institute and UCSF)

2021 **Teaching Assistant**, CS 154: Computer Architecture, UCSB

2019 **Teaching Assistant**, Research Mentorship Program (RMP), UCSB

2018–19 **NSF ERSP Mentor**, Department of Computer Science, UCSB
Students published PyRTLMatrix at the EMC2 workshop [W2]. Awarded the university-wide Fiona and Michael Goodchild Graduate Mentoring Award

2016–18 **Mentor**, Women in STEM Mentorship Program, UCSB

2015 **Teaching Assistant**, Department of Physics, UCSB

2015 **Mentor**, EUREKA Mentorship Program, California NanoSystems Institute, UCSB

Students mentored: Joann Chen (Research Intern, Meta, 2022), Manu Kondapaneni (2020), Junayed Naushad (2019), Dawit Aboye (2018–19), Dylan Kupsh (2018–19), Maggi Lim (2018–19), Jacqueline Mai (2018–19), Angela Yung (2016–17), Saurabh Gupta (2015)

Professional Service

2025 **CRA Congressional Visit Day**, Washington State contingent. Represented CRA and Washington State in Washington, D.C., in meetings with Members of Congress on federal support for computing research

2025 Invited participant, CCC Future of AI Research in Industry

2025 Invited participant, CCC Computing Futures Symposium

2022 Invited participant, CCC Mechanism Design for Improving Hardware Security [A2]

2022–25 Organizer, Undergraduate Architecture Mentoring Workshop (uArch)

2018–20 Co-President, Women in Computer Science (WiCS), UCSB

2019–20 Graduate Representative for Faculty Recruitment,
Department of Computer Science, UCSB

2018–19 Graduate Representative for Diversity, Department of Computer Science, UCSB

2017–19 Grace Hopper Celebration of Women in Computing (attendee)

2017–18 CRA-W Grad Cohort (attendee)

Program committees: MICRO 2026, ISPASS 2026, HPCA 2024–2026, ASPLOS 2025, ISCA 2024, SEED 2024, HASP 2023

External review committees: MICRO 2024, YArch 2023 and 2025, ISCA 2022, SEED 2021, ASPLOS 2020 (artifact evaluation)